Dr. Ing. Nicolo' Brandizzi

AI Researcher
Research Scientist @ Fraunhofer IAIS



CONTACTS

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Nicolo' Brandizzi
Scholar
1995

EXPERIENCE

Programming

Python 12 year(s) ●●●●●
LaTeX 10 year(s) ●●●●●
HPC 7 year(s) ●●●●○
Bash 10 year(s) ●●●●○
HTML/CSS 3 year(s) ●●●○○

Artificial Intelligence

Deep Learning/ ML 10 year(s) ●●●●●
Reinforcement Learning 8 year(s) ●●●●●
NLP / LLMs 8 year(s) ●●●●●
Data Engineer 3 year(s) ●●●○○
AI Governance 2 year(s) ●●○○○

Languages

Italian	Native
English	C2
German	B2
Spanish	B1

INTERESTS

Drum Playing | SCP | Mental Health | Neuro-diversity | Biking | Fermentation | Social Sciences

WORK EXPERIENCE

Spec Kitty – Head of Data 09/2026–Present

Responsible for evaluations of agentic coding, harnesses, and system data. Benchmark the models and the harnesses. Develop an environment for evolving existing harnesses based on evaluation results.

Fraunhofer IAIS – Research Scientist 03/2024–Present

Engineered HPC-scale data pipelines for LLM training (703 TB, 24 languages) within OpenGPT-X and EuroLingua GPT. Coordinated GDPR-compliant data governance in TrustLLM and Open-FLaaS. Lead reproducibility activities in DKZ.2R (data reproducibility); organized hackathons and provided consultancy on reproducible AI pipelines. Lead the Fraunhofer contribution to JUPITER AI Factory (EuroHPC, €25 M). Designed and coordinated evaluation campaigns for foundation models in ARCHER. Co-lead the Data Team (since 05/2025) and serve as Fraunhofer referent to the Lamarr Institute.

Sapienza University of Rome – Researcher 02/2023–03/2024

Research on human-AI interaction using hierarchical, model-based RL with communication models ("Trust Your Agent Project"). Co-authored publications, presented results at international workshops. Authored technical report on "Multi-Agent Model-Based RL" for the MBDA project, applying RL to mission-planning scenarios.

Basis Health – Research Scientist (Remote, USA) 10/2022–04/2023

Reviewed academic literature on wearable-tech innovation; advised R&D strategy. Developed and implemented health-data analysis methods enabling personalized insights. Produced actionable research reports for product teams.

Sapienza University of Rome – Teaching Assistant 07/2020–09/2022

Assisted in *Tecniche di Programmazione* and *Seminars in AI*; led tutorials, guided assignments, and supported course material development.

EDUCATION

Sapienza University of Rome – Ph.D. in Engineering in Computer Science 10/2020–05/2024

Thesis: "Conversational Agents in Human-Machine Interaction: Reinforcement Learning and Theory of Mind in Language Modeling". Supervisors: L. Iocchi, R. Navigli. Research visits: 10 mo @ U. Bayreuth (Mobile Intelligent UIs) + 5 mo @ U. Amsterdam (ILLC). Member of Ro.Co.Co. Lab, DIAG Department.

Sapienza University of Rome – M.Sc. AI & Robotics 10/2017–08/2020

Thesis: "Reinforcement Lupus: Cooperation through Emergent Communication in the Werewolf Game". Supervisor: L. Iocchi. Grade: 110 cum laude / 110 (GPA 4.0). Erasmus: 10 mo @ U. Groningen, supervisor D. Grossi.

Sapienza University of Rome – B.Sc. Computer & Autonomous Systems Engineering 10/2014–08/2017

Thesis: "Comparison Between Segmentation Methods". Supervisor: F. Pirri. Grade: 101 / 110 (GPA 2.97). Internship: 7 mo @ ALCOR Lab under L. Freda, working on the TRADR EU Project.

CERTIFICATIONS, GRANTS & AWARDS

Grants & Fellowships

EuroHPC JU – JUPITER AI Factory NRW €25 M • Proposal Co-author • Grant No. EUROHPC-2024-CEI-AI-01 • Pan-European AI Factory advancing Exascale AI research and infrastructure. 2025

AFOSR – Trust Your Agents €120 k • Research Collaborator • Grant FA8655-23-1-7257 • Human–machine cooperation via model-based and communication-augmented RL. 2023–2025

RWTH Aachen University – "Future of AI" Summit Participant • Expert forum bridging academia and industry on AI strategy and governance. 09/2023

University of Amsterdam – Research Software Engineering Grant Recipient • UvA Data Science Center grant supporting CorpusCompass development and data infrastructure optimization. 06/2023

Other Grants €15 972 in travel/research grants from TAILOR, COST, CINI, Sapienza.

Awards

Spotlight Paper – ALA21 Workshop @ AAMAS Selected as spotlight paper, recognizing excellence in research on adaptive and learning agents. 2021

Ph.D. Scholarship – Sapienza University of Rome First classified for competitive doctoral scholarship in Engineering in Computer Science. 2020

Certifications & Trainings

Fraunhofer – Project Management Training (16 h) Internal course on project coordination, budgeting, and EU project reporting. 03/2025

Fraunhofer – Leadership Program (7.5 h) Workshop on team motivation, conflict management, and communication. 02/2025

BlueDot Impact – AI Safety Fundamentals: AI Governance (3 mo) Online program on AI governance, risk assessment, and global coordination. 08/2024

CONFERENCES, SUMMER SCHOOLS & WORKSHOPS

Organized & Chaired

MULTITTRUST 5.0 5th International Workshop on Multidisciplinary Perspectives on Human-AI Team Trust 07/2026

AI-Mind Workshop 2.0 R2Net AI Research & Mental Well-Being Workshop at ESSAI2024 07/2024

MultiTTrust Special Issue at Journal of Interaction Studies 04/2024

MultiTTrust Workshop Multidisciplinary Perspectives on Human-AI Team Trust 12/2023

AI-Mind Workshop R2Net AI Research & Mental Well-Being Workshop at ESSAI2023 07/2023

Weights and Biases Workshop Workshop at Sapienza University 12/2022

AI4UAM PhD School Artificial Intelligence for Urban Air Mobility 09/2022

Invited Talks

European Broadcasting Union Data Technology Seminar 03/2025

AI-Mind Workshop 2.0 Advancing Generative and Trustworthy AI: Key Projects at Fraunhofer IAIS. 07/2024

Solarpunk Conference Federated Approaches to Data Challenges in Ethical AI 07/2024

4th TAILOR Conference Trustworthy AI from lab to market 06/2024

CLAIRE Rome Office Launch Safeguarding the Future: Higher Level AI Education 10/2023

AIhub New voices in AI 05/2022

Lightning Talks CLAIRE Rising Researchers Network 05/2022

Sapienza University of Rome Machine Learning Seminars 01/2021

PROFESSIONAL EXPERIENCES

IRIB-CNR – Institute for Biomedical Research, Italy 07/2024–Present

Co-investigator in the project "Stress-Related Difficulties and Neurodiversity Traits in People Working in AI" funded by the Ministry of Enterprises and Made in Italy (MIMIT). Designed and launched a large-scale survey (>500 AI professionals) on mental-health challenges in research.

Bayreuth University, Germany 11/2021–03/2025

Lead developer of CorpusCompass within the DFG-funded project "Modernity, Migration and Minorities". Secured the UvA Research Software Engineering Grant (2023). Published at CLiC-it 2023 and Field Matters 2024, ACL. Supervised two M.Sc. students; both continued as research assistants.

Sapienza University of Rome – PhD Student Representative 11/2021–10/2023

Advocated for PhD students, negotiated AI writing software licenses benefiting 30+ researchers, and contributed to departmental policy reforms.

DIAG, Sapienza University of Rome – Department Council Member 06/2021–10/2023

Collaborated on policy decisions to enhance research infrastructure and academic resources.

Rotaract Club, Rome 07/2021–07/2022

Developed and coordinated the Save Your Life project — a breast-cancer awareness initiative combining web outreach and community events.

COMMUNITY ENGAGEMENT

Steering & Expert Roles

MultiTTrust Steering Committee 11/2024–Present

Member of the international workshop series on Human-AI Team Trust; shaping themes, reviewing submissions, and fostering interdisciplinary collaboration.

Z-Inspection® Affiliate 11/2024–Present

Contribute to responsible-AI evaluation frameworks aligned with human-rights and ethical standards.

CAIRNE SIG – Ethical, Legal & Social Aspects of AI 10/2024–Present

Engaged in ethical-AI discussions, workshops on legal compliance, and integration of ELS principles into European projects.

CAIRNE R2Net Kernel Member 01/2022–Present

Coordinate 300-member network; oversee events, awards, and mental-well-being initiatives with ESSAI Summer School.

Volunteering & Student Initiatives

ADI – Associazione Dottorandi in Italia 11/2021–01/2024

Supported national initiatives improving doctoral research conditions in Italy.

Rotaract Club RTMM 04/2018–01/2024

Informatics consultant for social-impact projects; Internal Commission President (2017–2018) managing member coordination and events.

Erasmus Student Network Rome 03/2021–10/2022

Organized tandem-language exchanges and intercultural events (200+ participants each).

Conferences & Mentoring

HHAI2023 06/2023

Conference on Hybrid Human-Artificial Intelligence.

Virgilio Project 11/2021

Mentoring for Young Entrepreneurs and Start-ups.

PEER REVIEWS

Neurocomputing Elsevier journal focusing on computational neuroscience and neural systems. 2025

Autonomous Agents and Multi-Agent Systems Springer Nature journal on distributed AI and agent-based modeling. 2024

IEEE/RSJ IROS International Conference on Intelligent Robots and Systems. 2024

AI Communications IOS Press journal covering artificial intelligence theory and applications. 2023

EACL 17th European Chapter of the Association for Computational Linguistics (EACL 2023). 2023

CSAE 6th International Conference on Computer Science and Application Engineering. 2022

Emergent Communication Workshop – ICLR Workshop at the International Conference on Learning Representations. 2022

SUPERVISIONS

Master Students

Jonas Adler User-Centered Design of Digital Tools for Sociolinguistic Studies in Under-Resourced Languages – *Universität Bayreuth* 2024

Carsten Scholle User-Centered Design of Digital Tools for Sociolinguistic Studies in Under-Resourced Languages – *Universität Bayreuth* 2024

Luca Faraoni Multi-Agent Model-Based Reinforcement Learning – *Sapienza University of Rome* 2023

Abdolhadi Rezaei Hand Gesture and Voice Recognition on Raspberry Pi – *Sapienza University of Rome* 2022

Francesco Frattolillo Adaptive Playing and Opponent Modeling in Competitive Games – *Sapienza University of Rome* 2021

PUBLICATIONS

Journal Articles

[1] **N. Brandizzi**, M. E. Bailey, C. C. Jorge, M. C. Cohen, F. Frattolillo, A. R. Wagner, "Multidisciplinary perspectives on Human-AI team trust". In: *Interaction Studies* 26.2 (2025). Preface to the special issue, pp. 151–163. DOI: 10.1075/is.00025.edi. URL: https://doi.org/10.1075/is.00025.edi.

[2] **N. Brandizzi**, "Toward More Human-Like AI Communication: A Review of Emergent Communication Research". In: *IEEE Access* 11 (2023), pp. 142317–142340. DOI: 10.1109/ACCESS.2023.3339656.

[3] S. Pepe, S. Tedeschi, **N. Brandizzi**, S. Russo, L. Iocchi, C. Napoli, "Human Attention Assessment Using A Machine Learning Approach with GAN-based Data Augmentation Technique Trained Using a Custom Dataset". In: *OBM Neurobiology* 06.03 (Sept. 2022), p. 139. ISSN: 2573-4407. DOI: 10.21926/obm.neurobiol.2203139. URL: https://doi.org/10.21926/obm.neurobiol.2203139.

[4] **N. Brandizzi**, D. Grossi, L. Iocchi, "RLupus: Cooperation through emergent communication in The Werewolf social deduction game". In: *Intelligenza Artificiale* 15 (2021), pp. 55–70. ISSN: 2211-0097. DOI: 10.3233/IA-210081. URL: https://doi.org/10.3233/IA-210081.

Conference Papers

[5] M. Ali, M. Fromm, K. Thellmann, J. Ebert, A. A. Weber, R. Rutmann, C. Jain, M. Lübbering, D. Steinigen, J. Leveling, K. Klug, J. Schulze Buschhoff, L. Jurkschat, H. Abdelwahab, B. J. Stein, K.-H. Sylla, P. Denisov, **N. Brandizzi**, Q. Saleem, A. Bhowmick, L. Helmer, C. John, P. Ortiz Suarez, M. Ostendorff, A. Jude, L. Manjunath, S. Weinbach, C. Penke, O. Filatov, F. Barth, P. Mirza, L. Weber, I. Wendler, R. Sifa, F. Küch, A. Herten, R. Jäkel, G. Rehm, S. Kesselheim, J. Köhler, N. Flores-Herr, "Teuken-7B-base & Teuken-7b-instruct: Towards European LLMS". In: *Frontiers in Artificial Intelligence and Applications* (Oct. 2025). DOI: 10.3233/faia251328. URL: https://ebooks.iospress.nl/doi/10.3233/FAIA251328.

[6] G. Contino, R. Cipollone, F. Frattolillo, A. Fanti, **N. Brandizzi**, L. Iocchi, "Modeling a Trust Factor in Composite Tasks for Multi-Agent Reinforcement Learning". In: *Proceedings of the 12th International Conference on Human-Agent Interaction*. HAI '24. Swansea, United Kingdom: Association for Computing Machinery, 2024, pp. 195–203. DOI: 10.1145/3687272.3688325. URL: https://doi.org/10.1145/3687272.3688325.

[7] M. Adnan, **N. Brandizzi**, "CorpusCompass: A Tool for Data Extraction and Dataset Generation in Corpus Linguistics". In: *Proceedings of the 9th Italian Conference on Computational Linguistics*. Ed. by F. Boschetti, G. E. Lebani, B. Magnini, N. Novielli, vol. 3596. CEUR Workshop Proceedings. Venice, Italy: CEUR Workshop Proceedings, 2023, pp. 16–27. URL: http://ceur-ws.org/Vol-3596/.

[8] **N. Brandizzi**, D. Brunori, F. Frattolillo, A. Trapasso, L. Iocchi, "Mixed Human-UAV Reinforcement Learning: Literature Review and Open Challenges". In: *Proceedings of the 1st International Conference on Cognitive Aircraft Systems (ICCAS 2022)*. INSTICC. SciTePress, 2023, pp. 38–42. ISBN: 978-989-758-657-6. DOI: 10.5220/0011955100003622.

[9] **N. Brandizzi**, C. Centeio Jorge, R. Cipollone, F. Frattolillo, L. Iocchi, A.-S. Ulfert-Blank, "MULTITTRUST: 2nd Workshop on Multidisciplinary Perspectives on Human-AI Team Trust". In: *Proceedings of the 11th International Conference on Human-Agent Interaction*. HAI '23. Available online at https://multittrust.github.io/2ed/. Gothenburg,Sweden: Association for Computing Machinery, 2023, pp. 496–497. DOI: 10.1145/3623809.3623978. URL: https://doi.org/10.1145/3623809.3623978.

[10] E. Takmaz, **N. Brandizzi**, M. Giulianelli, S. Pezzelle, R. Fernández, "Speaking the Language of Your Listener: Audience-Aware Adaptation via Plug-and-Play Theory of Mind". In: *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*. Ed. by A. Rogers, J. L. Boyd-Graber, N. Okazaki, Association for Computational Linguistics, 2023, pp. 4198–4217. DOI: 10.18653/V1/2023.FINDINGS-ACL.258. URL: https://doi.org/10.18653/v1/2023.findings-acl.258.

[11] **N. Brandizzi**, A. Fanti, R. Gallotta, S. Russo, L. Iocchi, D. Nardi, C. Napoli, "Unsupervised Pose Estimation by Means of an Innovative Vision Transformer". In: *Artificial Intelligence and Soft Computing - 21st International Conference, ICAISC 2022, Zakopane, Poland, June 19-23, 2022, Proceedings, Part II*. Ed. by L. Rutkowski, R. Scherer, M. Korytkowski, W. Pedrycz, R. Tadeusiewicz, J. M. Zurada, vol. 13589. Lecture Notes in Computer Science. Springer, 2022, pp. 3–20. DOI: 10.1007/978-3-031-23480-4. URL: https://doi.org/10.1007/978-3-031-23480-4.

Workshop Papers

[12] J. Adler, C. Scholle, D. Buschek, **N. Brandizzi**, M. Adnan, "User-Centered Design of Digital Tools for Sociolinguistic Studies in Under-Resourced Languages". In: *Proceedings of the 3rd Workshop on NLP Applications to Field Linguistics (Field Matters 2024)*. Ed. by O. Serikov, E. Voloshina, A. Postnikova, S. Muradoglu, E. Le Ferrand, E. Klyachko, E. Vylomova, T. Shavrina, F. Tyers, Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 12–27. URL: https://aclanthology.org/2024.fieldmatters-1.3.

[13] **N. Brandizzi**, C. C. Jorge, F. Frattolillo, L. Iocchi, A. Ulfert-Blank, eds. *Proceedings of the 2nd International Workshop on Multidisciplinary Perspectives on Human-AI Team Trust co-located with 11th International Conference on Human-Agent Interaction (HAI 2023), Gothenburg, Sweden, December 4-7, 2023*. Vol. 3634. CEUR Workshop Proceedings. CEUR-WS.org, 2024. URL: https://ceur-ws.org/Vol-3634.

[14] S. Di Rienzo, F. Frattolillo, R. Cipollone, A. Fanti, **N. Brandizzi**, L. Iocchi, "Developing Targeted Communication through a Trust Factor in Multi-Agent Reinforcement Learning". In: *Proceedings of HHAI 2024 Workshops*. Ed. by P. Ericson, N. Khairova, M. D. Vos, vol. 3825. CEUR Workshop Proceedings. Published under Creative Commons License Attribution 4.0 International (CC BY 4.0). Malmö, Sweden: CEUR-WS.org, June 2024, pp. 171–175. URL: http://ceur-ws.org/Vol-3825/.

[15] F. Frattolillo, **N. Brandizzi**, R. Cipollone, L. Iocchi, "Towards Computational Models for Reinforcement Learning in Human-AI Teams". In: *Proceedings of the 2nd International Workshop on Multidisciplinary Perspectives on Human-AI Team Trust*. Vol. 3634. CEUR Workshop Proceedings. Gothenburg, Sweden, 2023, pp. 41–51. URL: https://ceur-ws.org/Vol-3634/.

[16] **N. Brandizzi**, L. Iocchi, "Emergent communication in human-machine games". In: *Emergent Communication Workshop at ICLR 2022*. 2022. URL: https://openreview.net/forum?id=rqLgeQWCXZ9.

[17] R. Aureli, **N. Brandizzi**, G. De Magistris, R. Brociek, "A Customized Approach to Anomalies Detection by using Autoencoders". In: *Proceedings of the Scholar's Yearly Symposium of Technology, Engineering and Mathematics (SYSTEM 2021), Catania, Italy, July 27-29, 2021*. Ed. by R. Fazzolari, A. A. Jaber, vol. 3092. CEUR Workshop Proceedings. CEUR-WS.org, 2021, pp. 53–59. URL: http://ceur-ws.org/Vol-3092/p09.pdf.

[18] **N. Brandizzi**, V. Bianco, G. Castro, S. Russo, A. Wajda, "Automatic RGB Inference Based on Facial Emotion Recognition". In: *Proceedings of the Scholar's Yearly Symposium of Technology, Engineering and Mathematics (SYSTEM 2021), Catania, Italy, July 27-29, 2021*. Ed. by R. Fazzolari, A. A. Jaber, vol. 3092. CEUR Workshop Proceedings. CEUR-WS.org, 2021, pp. 66–74. URL: http://ceur-ws.org/Vol-3092/p11.pdf.

[19] **N. Brandizzi**, A. Fanti, R. Gallotta, C. Napoli, "FEFFuL: a Few-Examples Fitness Function Learner". In: *Proceedings of the Scholar's Yearly Symposium of Technology, Engineering and Mathematics (SYSTEM 2021), Catania, Italy, July 27-29, 2021*. Ed. by R. Fazzolari, A. A. Jaber, vol. 3092. CEUR Workshop Proceedings. CEUR-WS.org, 2021, pp. 75–79. URL: http://ceur-ws.org/Vol-3092/p12.pdf.

[20] **N. Brandizzi**, S. Russo, R. Brociek, A. Wajda, "First Studies to Apply the Theory of Mind Theory to Green and Smart Mobility by Using Gaussian Area Clustering". In: *Proceedings of the International Conference of Yearly Reports on Informatics, Mathematics and Engineering, Online, July 9, 2021*. Ed. by R. Fazzolari, A. A. Jaber, vol. 3118. CEUR Workshop Proceedings. CEUR-WS.org, 2021, pp. 71–76. URL: http://ceur-ws.org/Vol-3118/p10.pdf.

Preprints

[21] **N. Brandizzi**, "Extended Abstract: Conversational Agents in Human-Machine Interaction, Reinforcement Learning and Theory of Mind in Language Modeling". In: May 2024, p. 4. DOI: 10.13140/RG.2.2.29722.25287. URL: https://www.researchgate.net/publication/380396634_Extended_Abstract_Conversational_Agents_in_Human-Machine_Interaction_Reinforcement_Learning_and_Theory_of_Mind_in_Language_Modeling.

[22] **N. Brandizzi**, H. Abdelwahab, A. Bhowmick, L. Helmer, B. J. Stein, P. Denisov, Q. Saleem, M. Fromm, M. Ali, R. Rutmann, F. Naderi, M. S. Agy, A. Schwirjow, F. Küch, L. Hahn, M. Ostendorff, P. O. Suarez, G. Rehm, D. Wegener, N. Flores-Herr, J. Köhler, J. Leveling, *Data Processing for the OpenGPT-X Model Family*. Oct. 2024. arXiv: 2410.08800 [cs.CL]. URL: https://arxiv.org/abs/2410.08800.

Books

[23] **N. Brandizzi**, M. E. Bailey, C. C. Jorge, M. C. Cohen, F. Frattolillo, A. R. Wagner, eds. *Multidisciplinary Perspectives on Human-AI Team Trust*. Interaction Studies 26.2 (2025). Special issue. DOI: 10.1075/is.26.2. URL: https://doi.org/10.1075/is.26.2.

Other

[24] **N. Brandizzi**, "Conversational Agents in Human-Machine Interaction: Reinforcement Learning and Theory of Mind in Language Modeling". PhD thesis. 2024. URL: https://hdl.handle.net/11573/1709118.